MACHINE LEARNING-BASED ANALYSIS OF HOUSEHOLD EXPENDITURE PATTERNS

A Study of Food and Transportation Expenditure in Urban Households of Maharashtra

PROJECT REPORT

Machine Learning | Data Analysis | Predictive Modelling

Dataset: Maharashtra Household Expenditure Data
Total observations: 10,567

ABSTRACT

Household expenditure varies according to socioeconomic and demographic characteristics. Understanding these variations can provide insights into consumption patterns and household financial behaviour.

This project applies data analysis and machine learning techniques to study monthly food and transportation expenditure among households in Maharashtra. The analysis examines the relationship between expenditure and factors such as household income, district, household size, age, occupation, education and gender.

Multiple machine learning approaches are used to predict household expenditure. Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Feature importance and model error analysis are also used to interpret predictive performance.

1. INTRODUCTION

Household expenditure is an important indicator of economic behaviour and living standards. Food and transportation are two major components of monthly household expenditure.

Expenditure patterns may differ across geographical regions and socioeconomic groups. Machine learning provides methods for identifying complex relationships between household characteristics and expenditure.

The present study focuses on identifying expenditure patterns and developing predictive models for food and transportation expenditure in Maharashtra.

2. PROBLEM STATEMENT

The objective is to analyse household food and transportation expenditure patterns across socioeconomic and demographic characteristics in Maharashtra.

The project investigates whether household income, district, household size, age, occupation, education and gender can explain variations in monthly expenditure.

3. OBJECTIVES

1. To analyse household food and transportation expenditure.

2. To examine relationships between income and expenditure.

3. To identify important socioeconomic predictors.

4. To develop machine learning models for expenditure prediction.

5. To compare predictive performance across models.

6. To interpret model behaviour using feature importance and SHAP.

7. To investigate prediction errors and district-wise variations.

4. DATASET DESCRIPTION

The cleaned dataset contains 10,567 household observations and 12 variables.

Variable	Description
HH_ID	Unique household identifier
STATE	State
DISTRICT	District
REGION_TYPE	Region classification
AGE_GROUP	Household age category
OCCUPATION_GROUP	Occupation category
EDU_GROUP	Education category
GENDER_GROUP	Gender category
SIZE_GROUP	Household size category
M_EXP_FOOD	Monthly food expenditure
M_EXP_TRANSPORT	Monthly transportation expenditure
TOT_INCOME	Total household income

5. DATA PREPROCESSING AND CLEANING

Data preprocessing was performed before statistical analysis and model development. The cleaning workflow included checking missing values, duplicate household identifiers, numeric data types and invalid expenditure or income values.

Parameter	Result
Total observations	10567
Total variables	12
Missing values	0
Duplicate household IDs	0

6. EXPLORATORY DATA ANALYSIS

Descriptive Statistics

Statistic	TOT_INCOME	M_EXP_FOOD	M_EXP_TRANSPORT
count	10567.0	10567.0	10567.0
mean	56810.55	10310.97	991.78
std	39791.95	4077.44	1320.52
min	0.0	1670.0	0.0
25%	31500.0	7412.5	380.0

Statistic	TOT_INCOME	M_EXP_FOOD	M_EXP_TRANSPORT
50%	46000.0	9640.0	650.0
75%	71000.0	12500.0	1250.0
max	506000.0	39640.0	49150.0

7. CORRELATION ANALYSIS

Variable	TOT_INCOME	M_EXP_FOOD	M_EXP_TRANSPORT
TOT_INCOME	1.0000	0.4595	0.2948
M_EXP_FOOD	0.4595	1.0000	0.3970
M_EXP_TRANSPORT	0.2948	0.3970	1.0000

Pearson correlation measures the strength and direction of linear association between numerical variables. Correlation does not establish causation.

8. MACHINE LEARNING METHODOLOGY

The machine learning workflow used household socioeconomic and demographic characteristics as input features. Monthly food and transportation expenditure were treated as separate prediction targets.

Input Features

• TOT_INCOME

• DISTRICT

• AGE_GROUP

• OCCUPATION_GROUP

• EDU_GROUP

• GENDER_GROUP

• SIZE_GROUP

Prediction Targets

1. Monthly food expenditure (M_EXP_FOOD)

2. Monthly transportation expenditure (M_EXP_TRANSPORT)

Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 split. The training set contained 8,453 observations and the testing set contained 2,114 observations.

9. MODEL PERFORMANCE EVALUATION

Model_Error_Performance.csv

Model	Target	MAE	MSE	RMSE	R2_Score
Food	M_EXP_FOOD	1799.9737	7021772.1083	2649.8627	0.5816
Transportation	M_EXP_TRANSPORT	526.2764	1206392.2738	1098.3589	0.2542

MAE measures the average absolute prediction error. RMSE gives greater weight to larger errors. R^2 measures the proportion of target variance explained by the model.

10. FEATURE IMPORTANCE ANALYSIS

feature_importance.csv

Feature	Importance_Mean	Importance_Std	Target	Best_Model
DISTRICT	0.3412	0.0151	Transportation	Linear Regression
TOT_INCOME	0.0432	0.0052	Transportation	Linear Regression
SIZE_GROUP	0.0028	0.0012	Transportation	Linear Regression
EDU_GROUP	0.0017	0.0016	Transportation	Linear Regression
GENDER_GROUP	0.0010	0.0011	Transportation	Linear Regression
OCCUPATION_GROUP	0.0009	0.0006	Transportation	Linear Regression
AGE_GROUP	-0.0002	0.0008	Transportation	Linear Regression
DISTRICT	0.6187	0.0300	Food	Bagging Regressor
TOT_INCOME	0.1881	0.0117	Food	Bagging Regressor
SIZE_GROUP	0.1547	0.0124	Food	Bagging Regressor
AGE_GROUP	0.0100	0.0038	Food	Bagging Regressor
OCCUPATION_GROUP	0.0070	0.0040	Food	Bagging Regressor
EDU_GROUP	0.0037	0.0050	Food	Bagging Regressor
GENDER_GROUP	-0.0001	0.0056	Food	Bagging Regressor

SHAP_Feature_Importance.csv

Feature	Mean_Absolute_SHAP	Target
DISTRICT	1875.4980	M_EXP_FOOD
TOT_INCOME	1024.4145	M_EXP_FOOD
SIZE_GROUP	499.5552	M_EXP_FOOD
GENDER_GROUP	157.7194	M_EXP_FOOD
AGE_GROUP	146.9775	M_EXP_FOOD

Feature	Mean_Absolute_SHAP	Target
OCCUPATION_GROUP	134.7486	M_EXP_FOOD
EDU_GROUP	91.4777	M_EXP_FOOD
DISTRICT	419.2817	M_EXP_TRANSPORT
TOT_INCOME	122.4993	M_EXP_TRANSPORT
EDU_GROUP	54.1012	M_EXP_TRANSPORT
SIZE_GROUP	35.7165	M_EXP_TRANSPORT
GENDER_GROUP	32.8760	M_EXP_TRANSPORT
AGE_GROUP	25.5621	M_EXP_TRANSPORT
OCCUPATION_GROUP	20.2700	M_EXP_TRANSPORT

Feature importance analysis helps identify which input variables contribute most to predictive performance. Permutation importance measures the reduction in model performance when a feature is shuffled.

11. SHAP EXPLAINABILITY ANALYSIS

SHAP_Feature_Importance.csv

Feature	Mean_Absolute_SHAP	Target
DISTRICT	1875.4980	M_EXP_FOOD
TOT_INCOME	1024.4145	M_EXP_FOOD
SIZE_GROUP	499.5552	M_EXP_FOOD
GENDER_GROUP	157.7194	M_EXP_FOOD
AGE_GROUP	146.9775	M_EXP_FOOD
OCCUPATION_GROUP	134.7486	M_EXP_FOOD
EDU_GROUP	91.4777	M_EXP_FOOD
DISTRICT	419.2817	M_EXP_TRANSPORT
TOT_INCOME	122.4993	M_EXP_TRANSPORT
EDU_GROUP	54.1012	M_EXP_TRANSPORT
SIZE_GROUP	35.7165	M_EXP_TRANSPORT
GENDER_GROUP	32.8760	M_EXP_TRANSPORT
AGE_GROUP	25.5621	M_EXP_TRANSPORT
OCCUPATION_GROUP	20.2700	M_EXP_TRANSPORT

SHAP (SHapley Additive exPlanations) is used to interpret individual model predictions by estimating the contribution of each feature to a prediction.

12. MODEL ERROR ANALYSIS

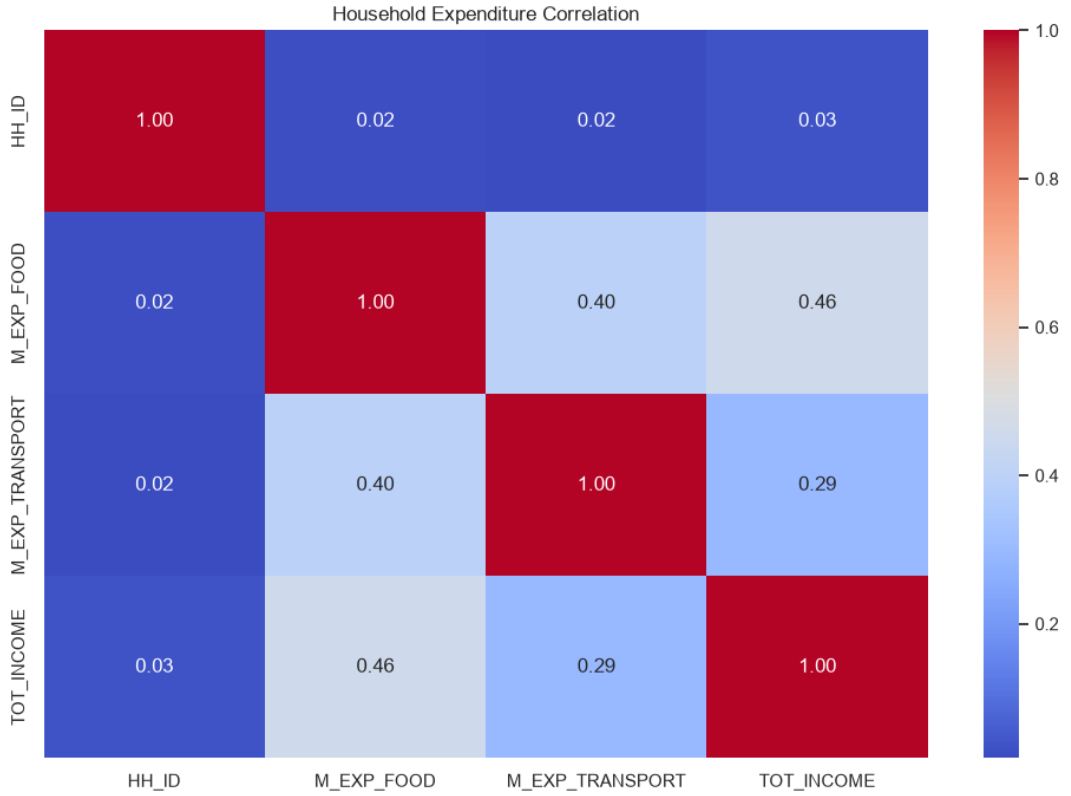
District-wise Prediction Errors

DISTRICT	MAE	Mean_Actual	Mean_Predicted	Household_Count	Target
Ahilyanagar	1507.4928	7156.2874	7216.2690	87	M_EXP_FOOD
Amravati	2170.9516	9042.8500	9634.5136	60	M_EXP_FOOD
Bhandara	1691.2060	10490.1667	10727.7124	30	M_EXP_FOOD
Bid	956.4917	6373.3333	7317.6050	3	M_EXP_FOOD
Buldana	936.4662	9187.2143	9364.5945	28	M_EXP_FOOD
Chandrapur	1891.4303	9493.2394	10517.1728	71	M_EXP_FOOD
Chhatrapati Sambhajinagar	701.8051	8316.6939	8306.9192	49	M_EXP_FOOD
Gondiya	1793.6881	11038.0435	10772.1385	46	M_EXP_FOOD
Jalgaon	880.2821	8630.9691	8279.5203	97	M_EXP_FOOD
Kolhapur	816.1445	7000.1532	7044.1331	124	M_EXP_FOOD
Mumbai	4164.8495	15191.2075	14058.4995	53	M_EXP_FOOD
Mumbai Suburban	3589.8079	14359.8675	14368.1259	166	M_EXP_FOOD
Nagpur	1144.9550	7963.0992	7938.2722	121	M_EXP_FOOD
Nanded	984.9420	8227.2105	8312.9347	57	M_EXP_FOOD
Nashik	925.7719	8249.1261	8294.1111	111	M_EXP_FOOD
Parbhani	1152.1089	7323.1562	7852.5797	32	M_EXP_FOOD
Pune	1473.0881	8508.9091	8422.7004	220	M_EXP_FOOD
Raigarh	3093.1091	16095.3256	16038.8879	43	M_EXP_FOOD
Ratnagiri	827.4761	12888.1250	12494.6414	16	M_EXP_FOOD
Satara	1970.8168	7946.2222	7653.3492	36	M_EXP_FOOD

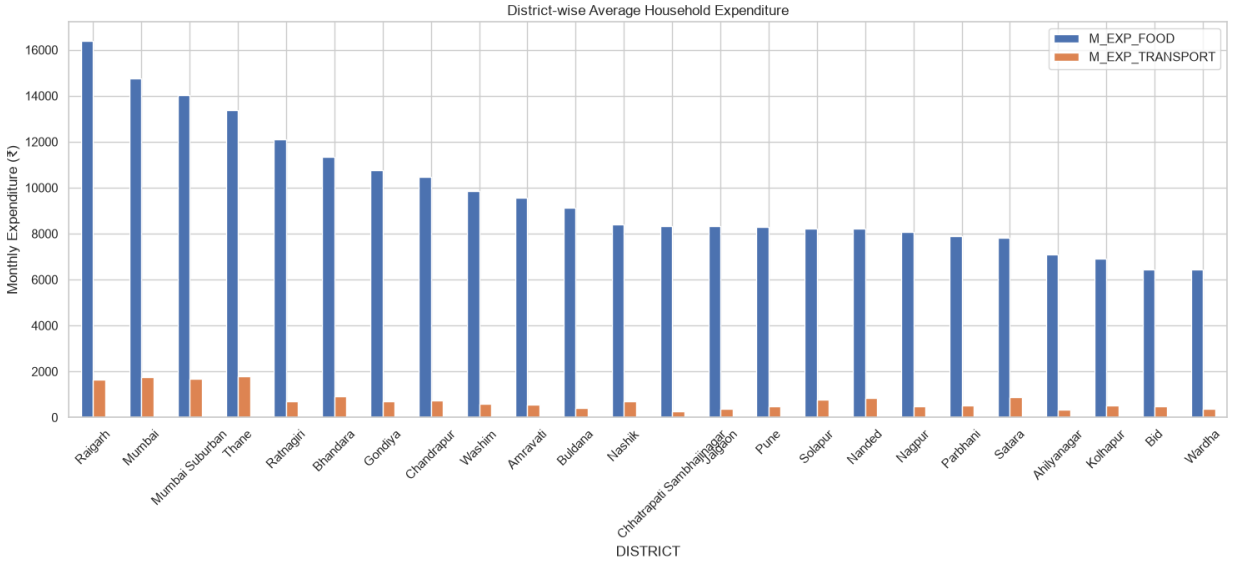
Residuals are calculated as the difference between actual and predicted expenditure. Residual analysis helps assess prediction bias, error distribution and model limitations.

13. VISUAL ANALYSIS AND GRAPHS

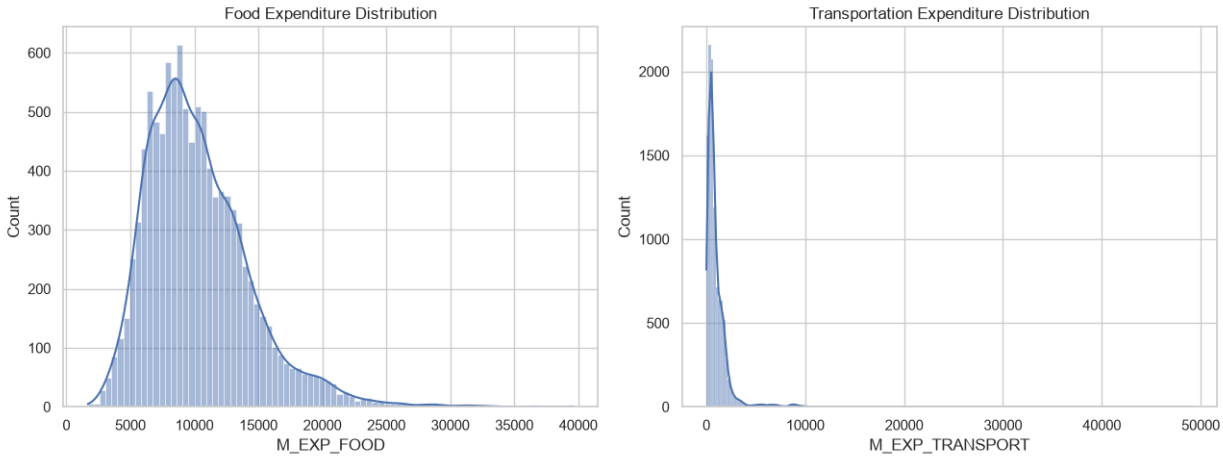
A total of 27 PNG visualisations were located in the project graph directories.



Correlation Heatmap



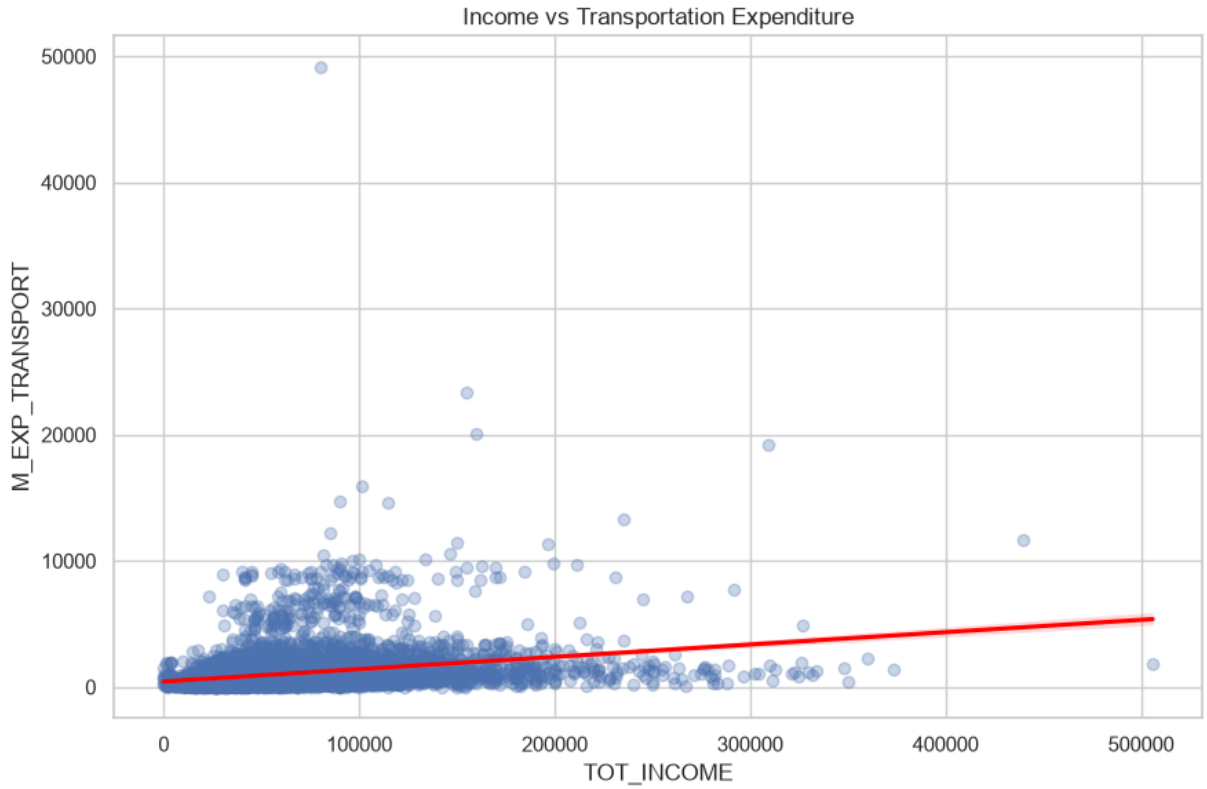
District Comparison



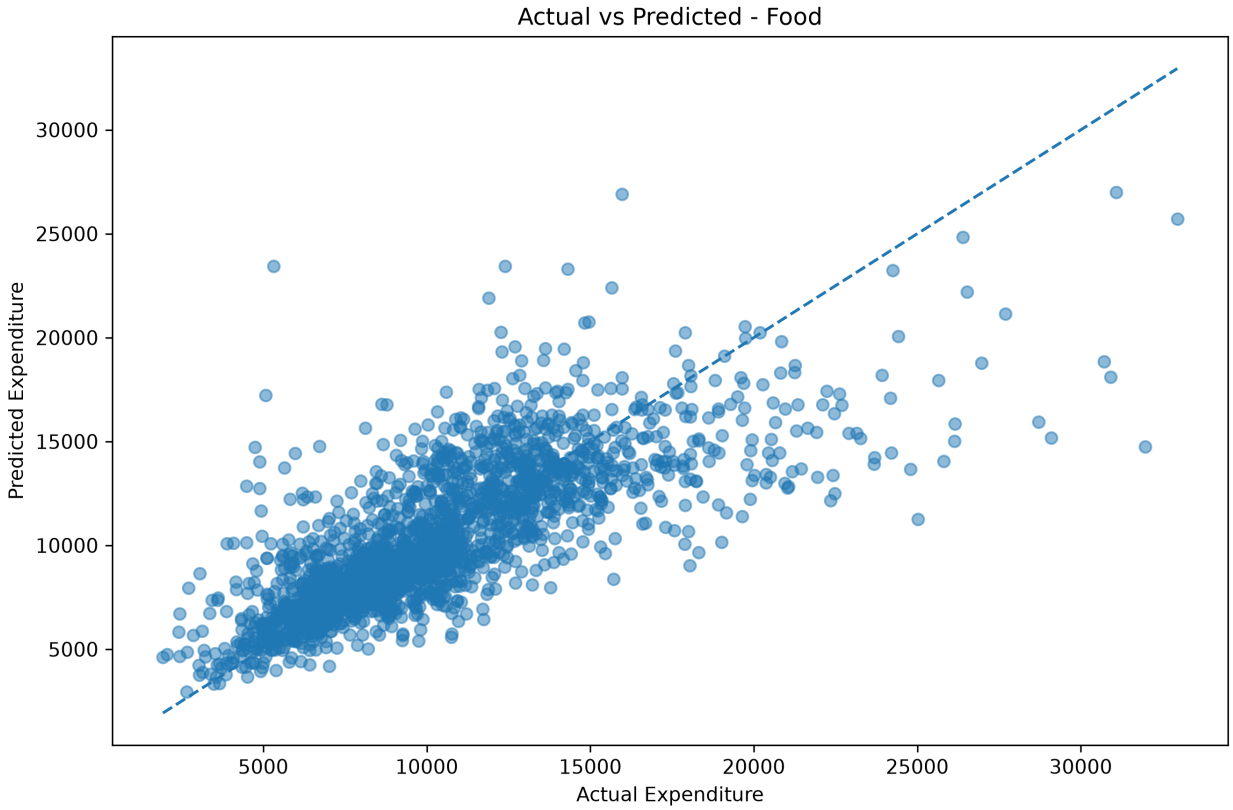
Expenditure Distribution



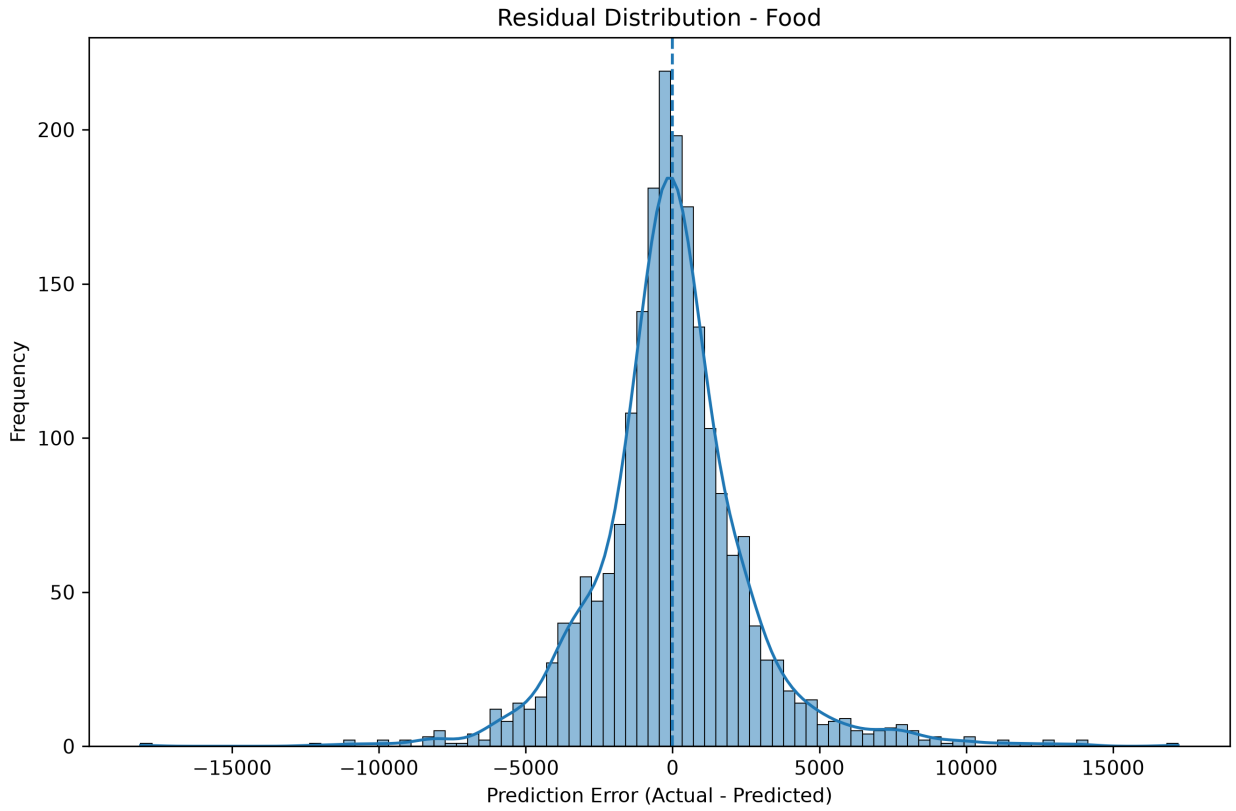
Income Vs Food



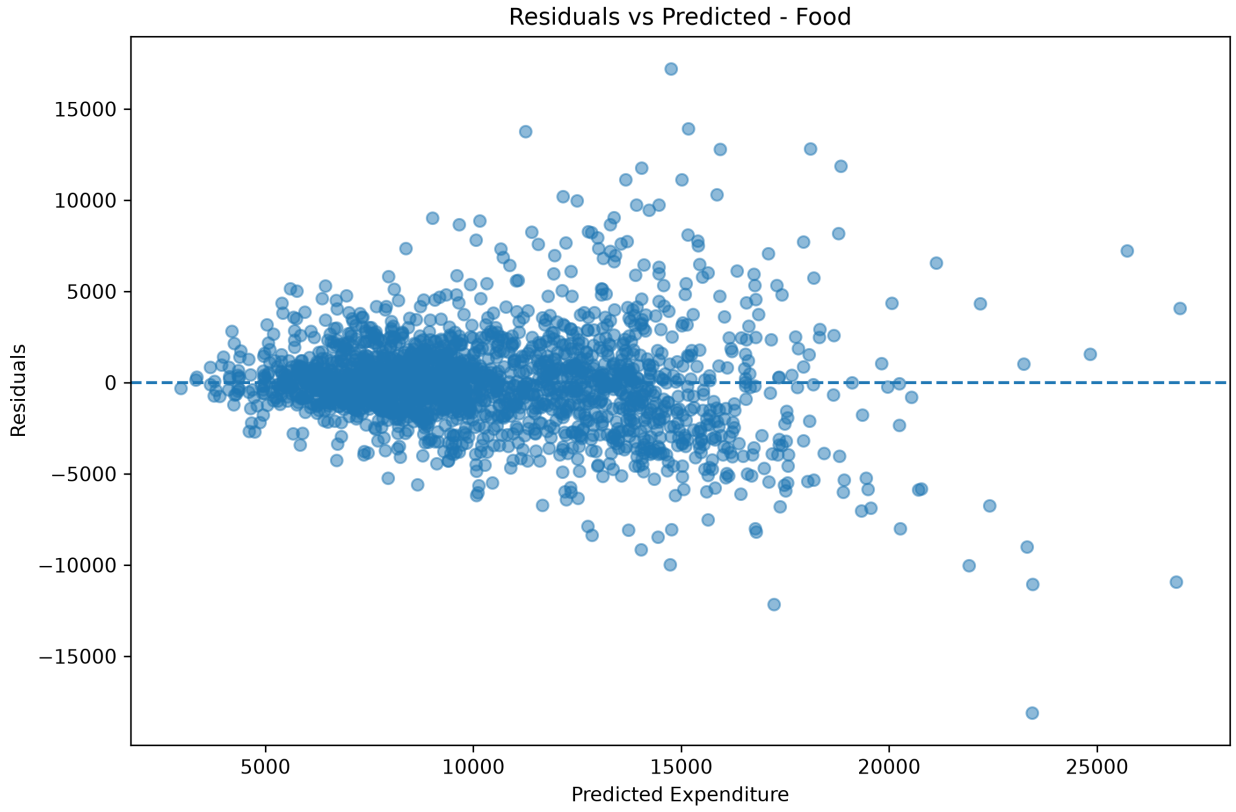
Income Vs Transport



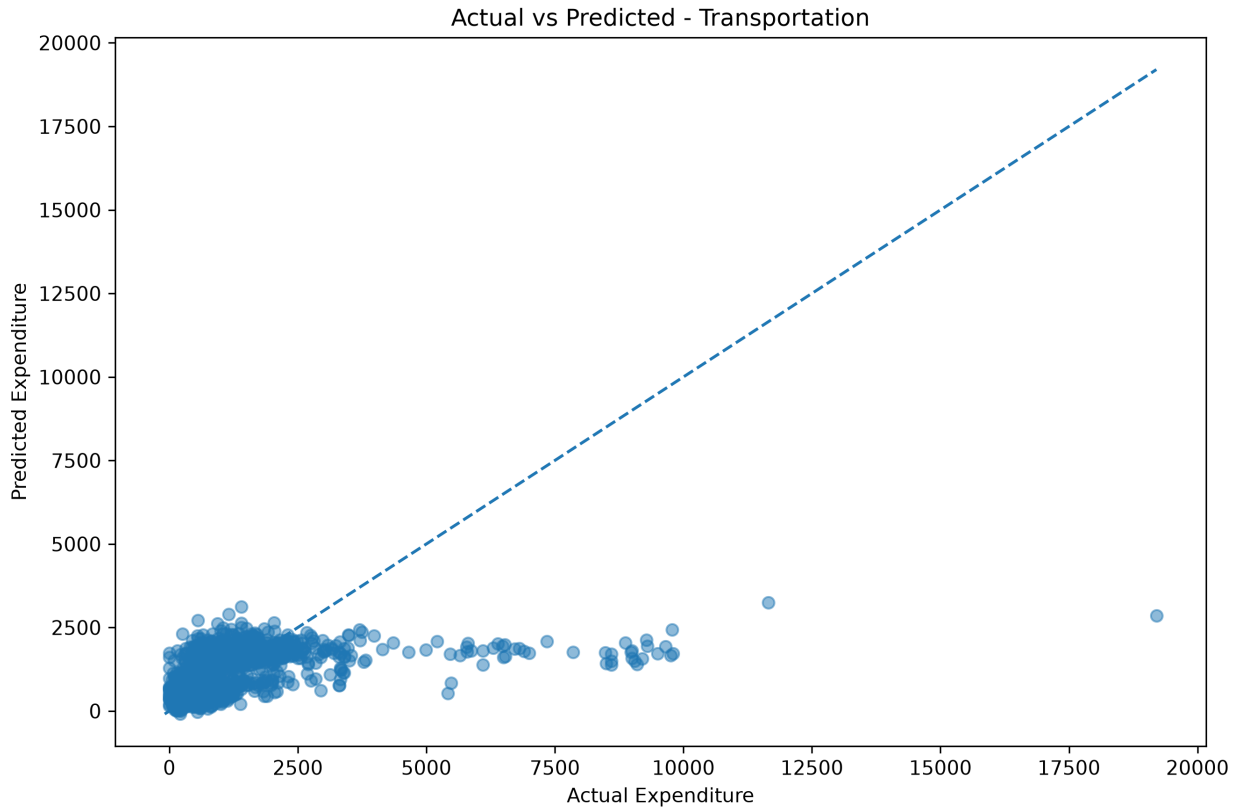
M Exp Food Actual Vs Predicted



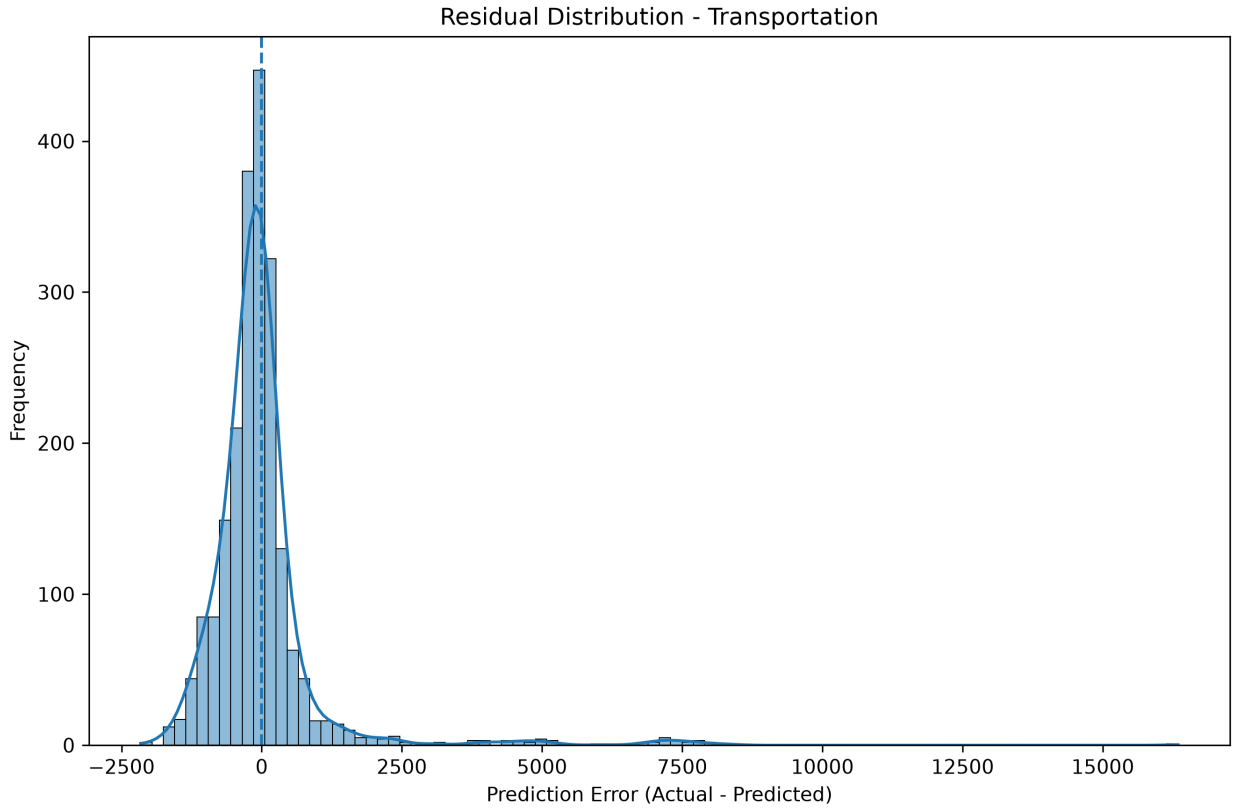
M Exp Food Residual Distribution



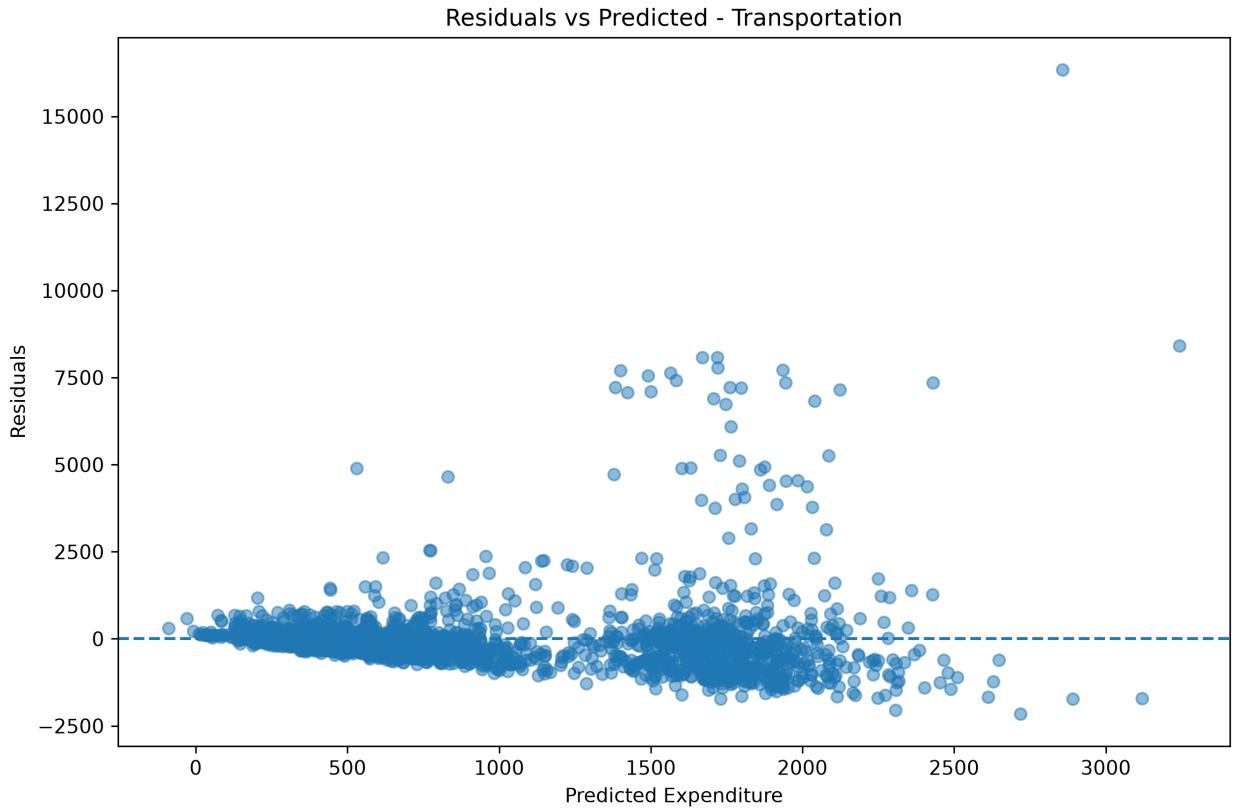
M Exp Food Residual Vs Predicted



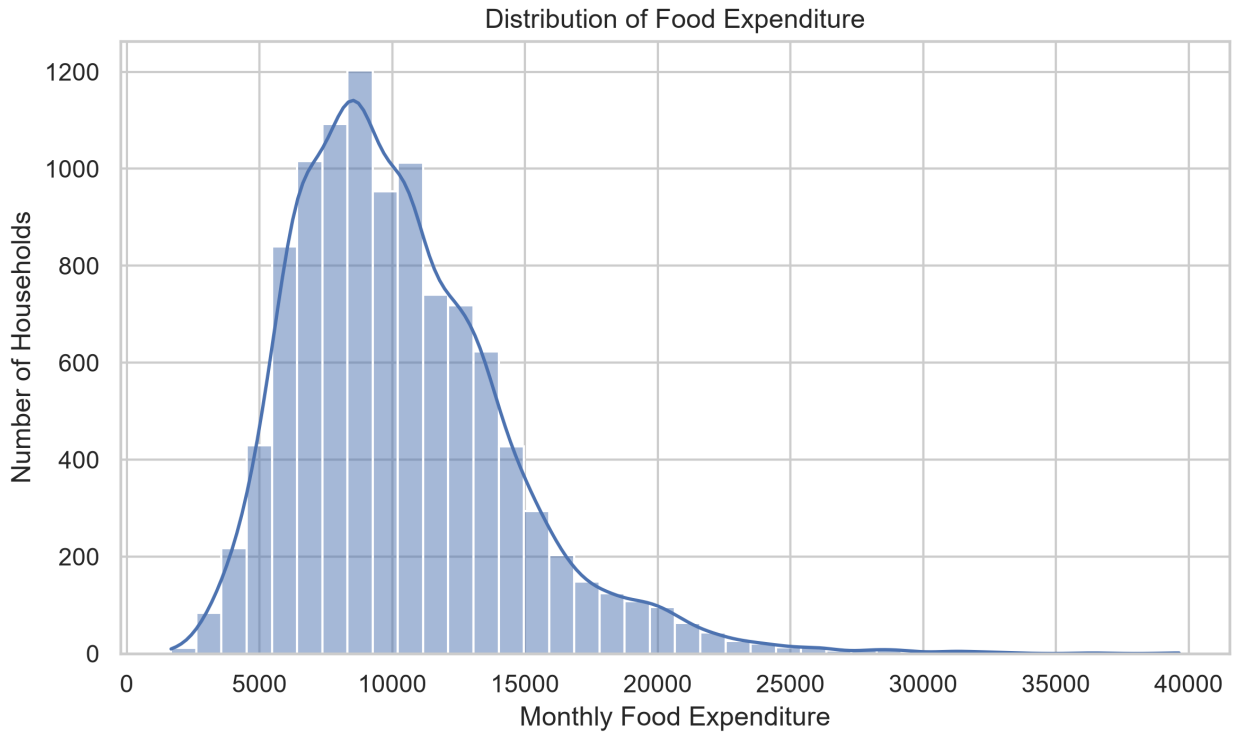
M Exp Transport Actual Vs Predicted



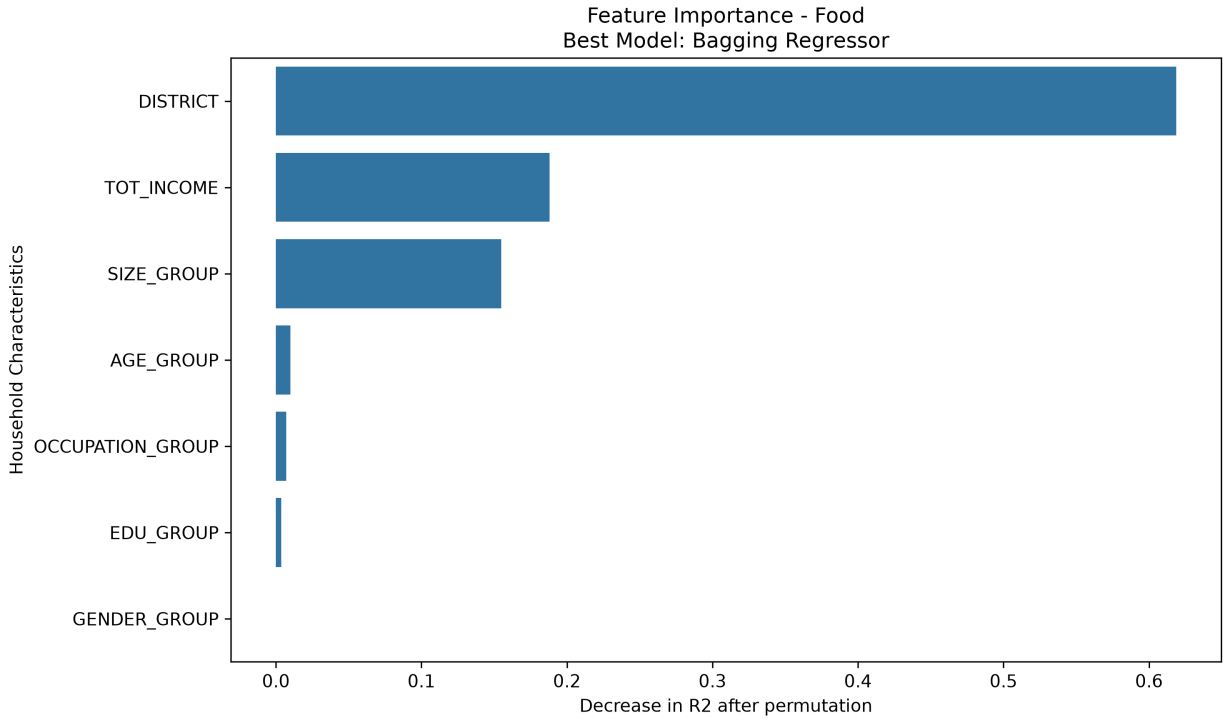
M Exp Transport Residual Distribution



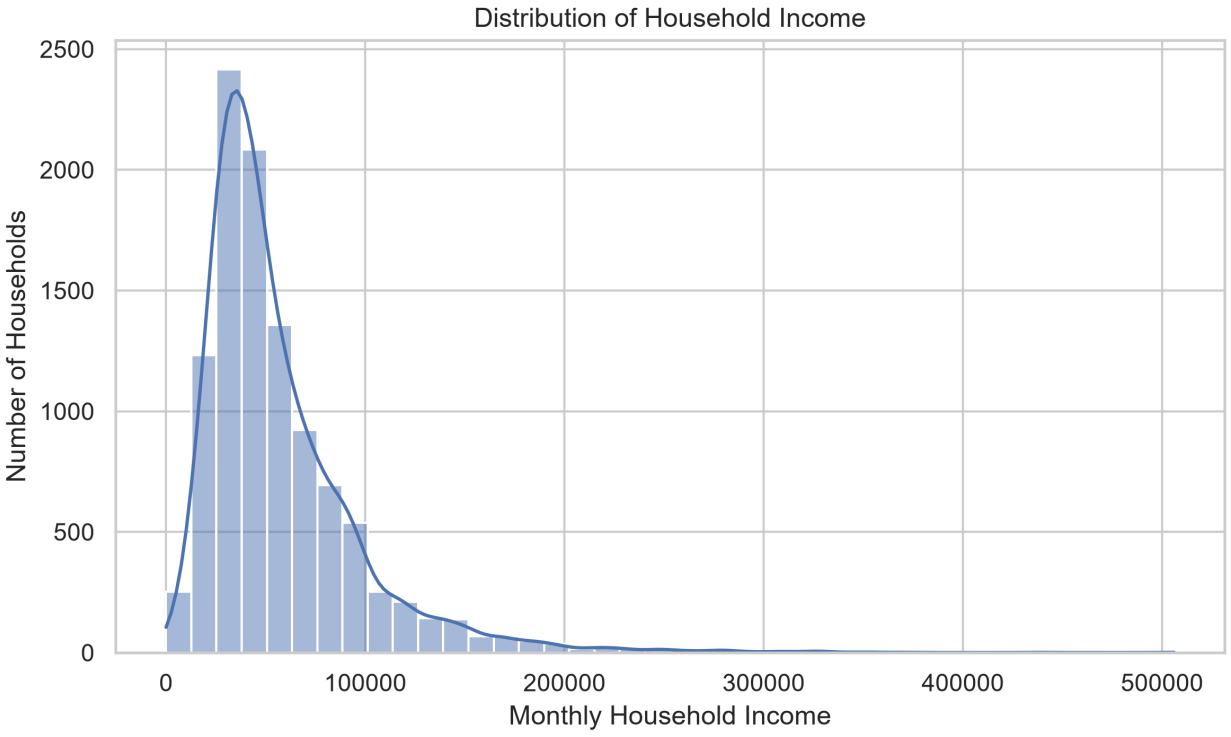
M Exp Transport Residual Vs Predicted



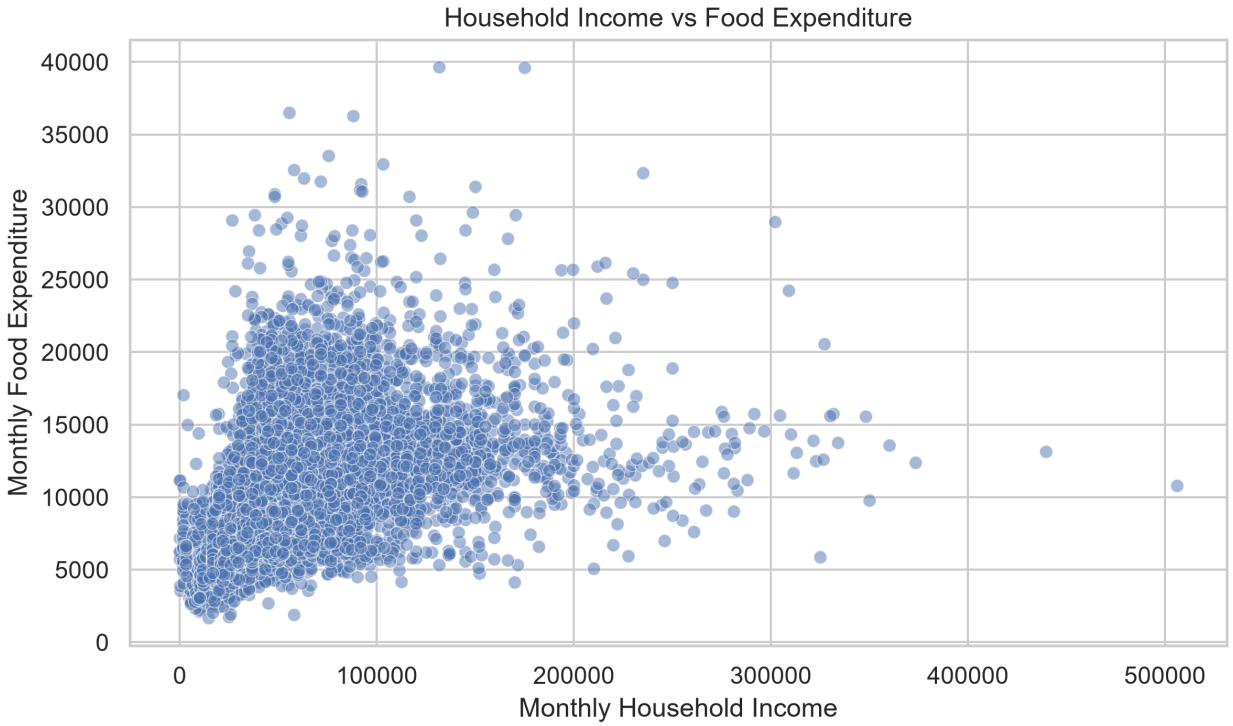
Food Distribution



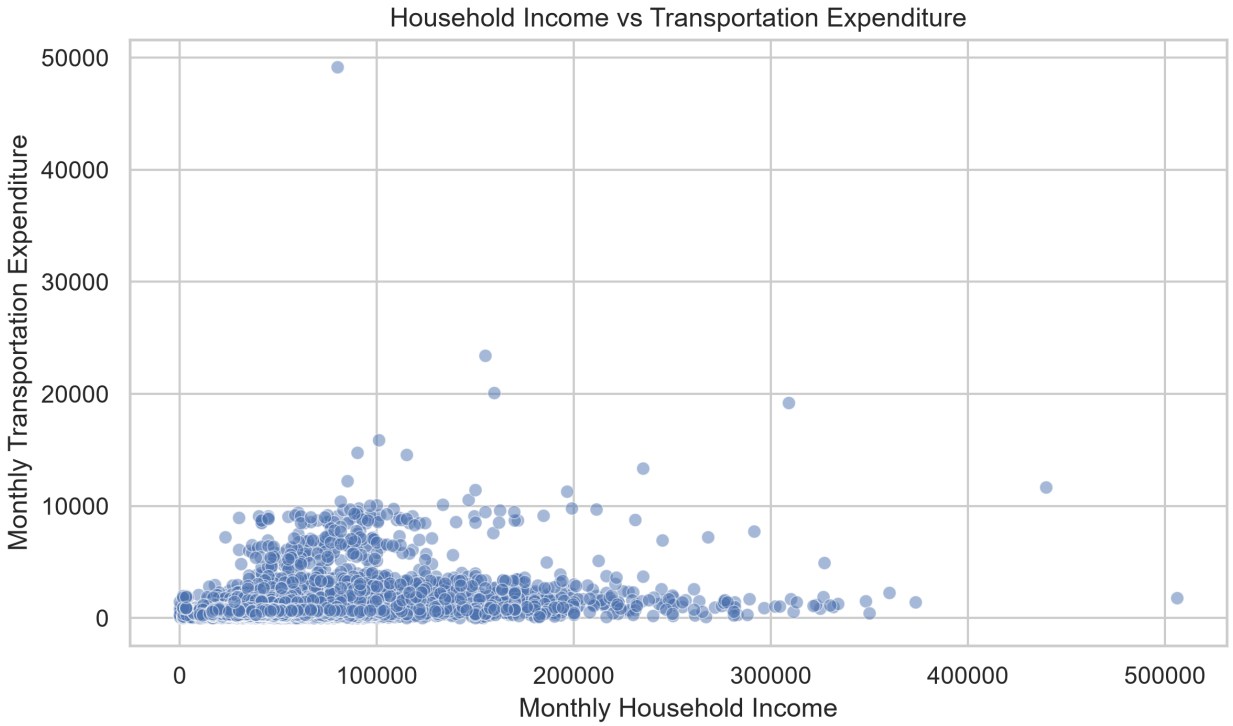
Food Feature Importance



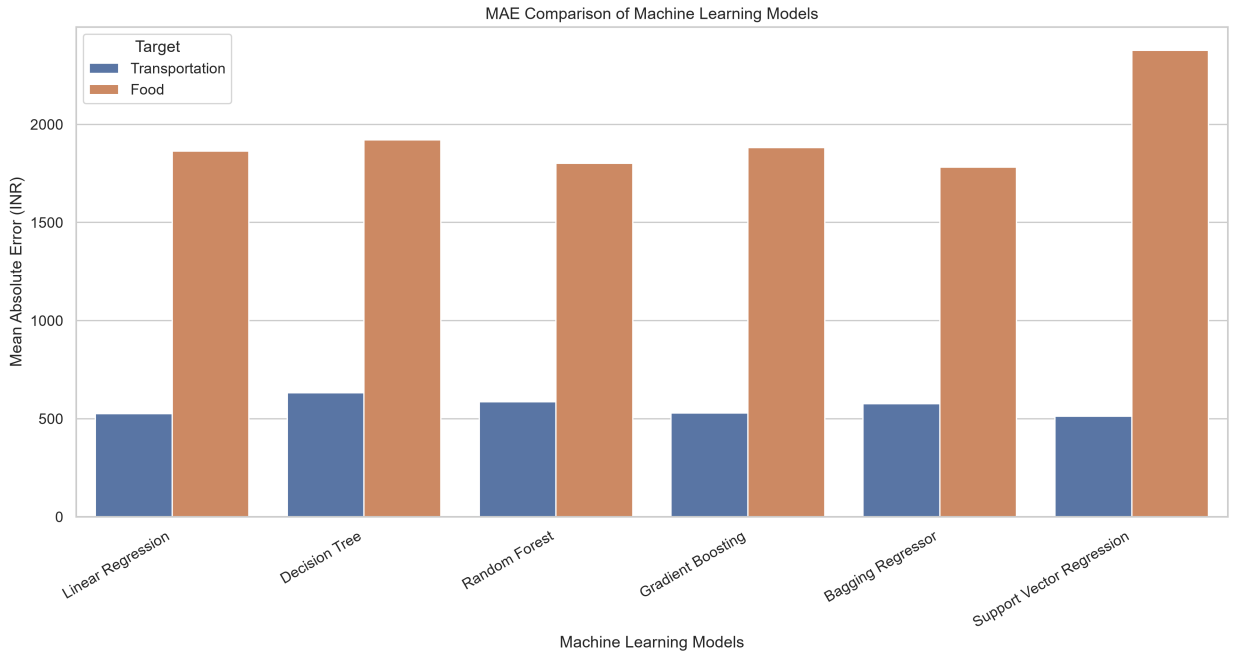
Income Distribution



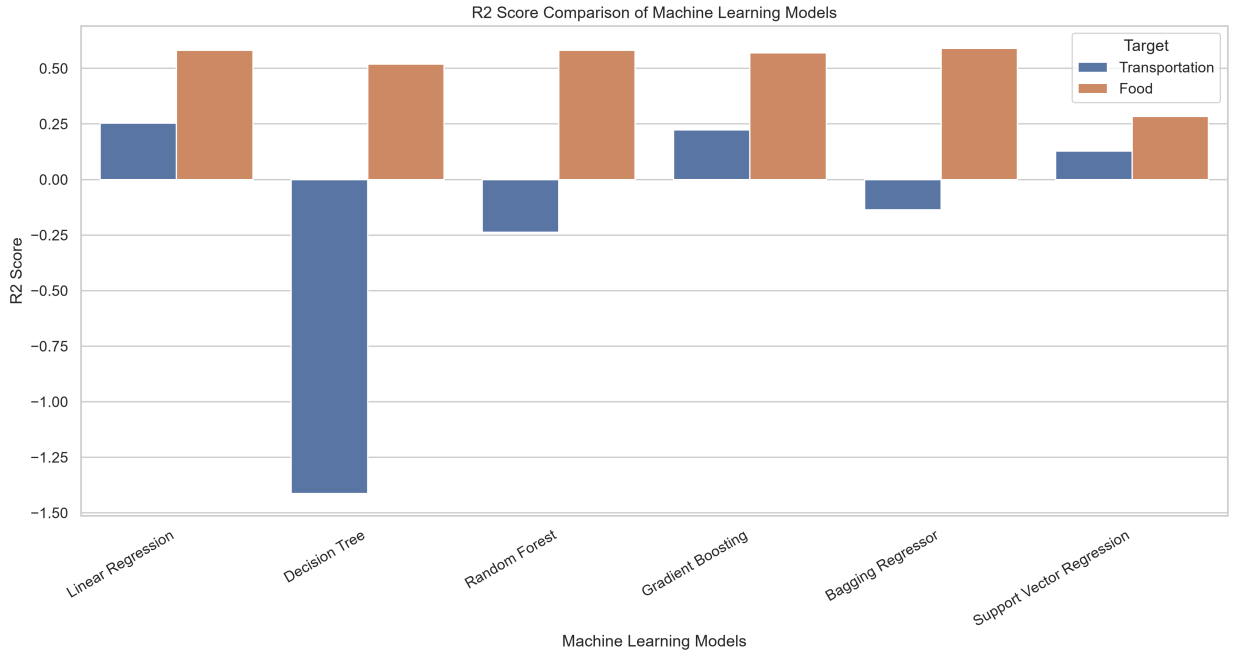
Income Vs Food



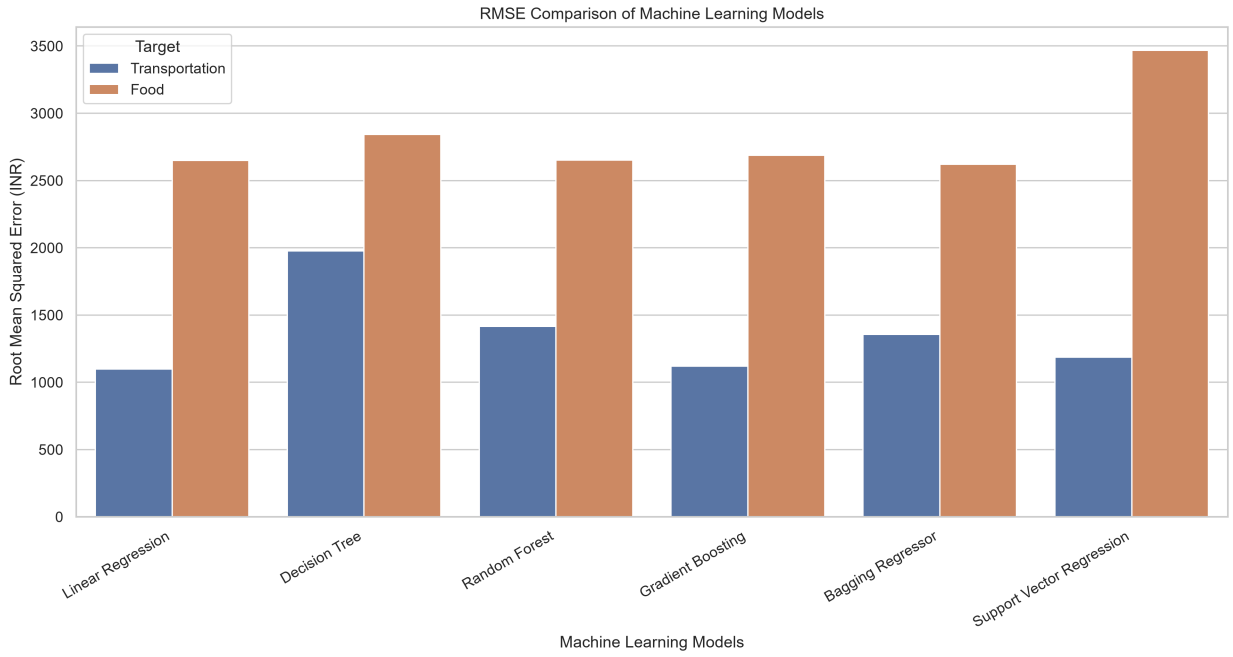
Income Vs Transport



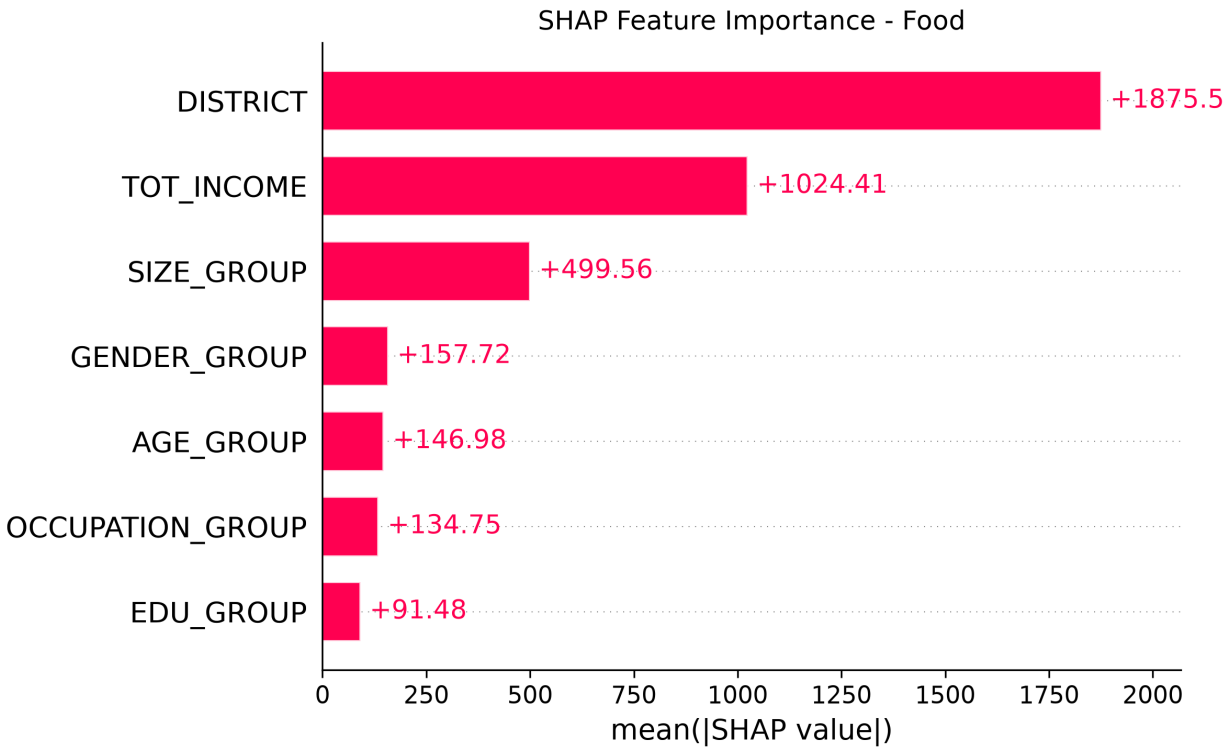
Mae Comparison



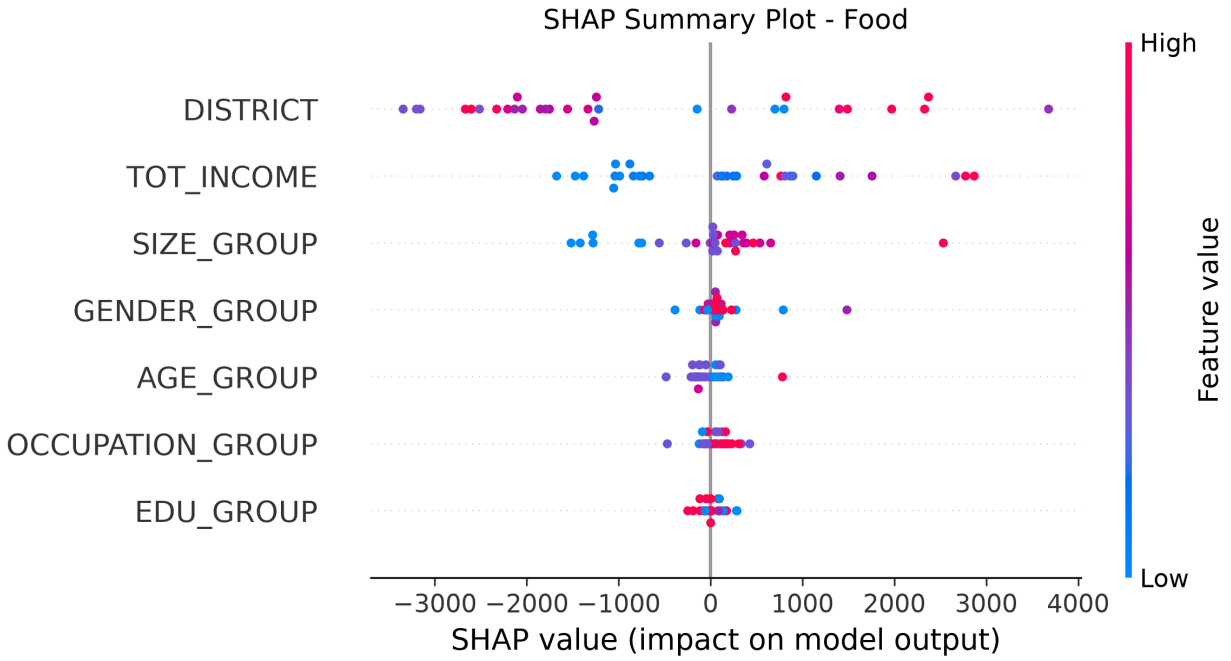
R2 Comparison



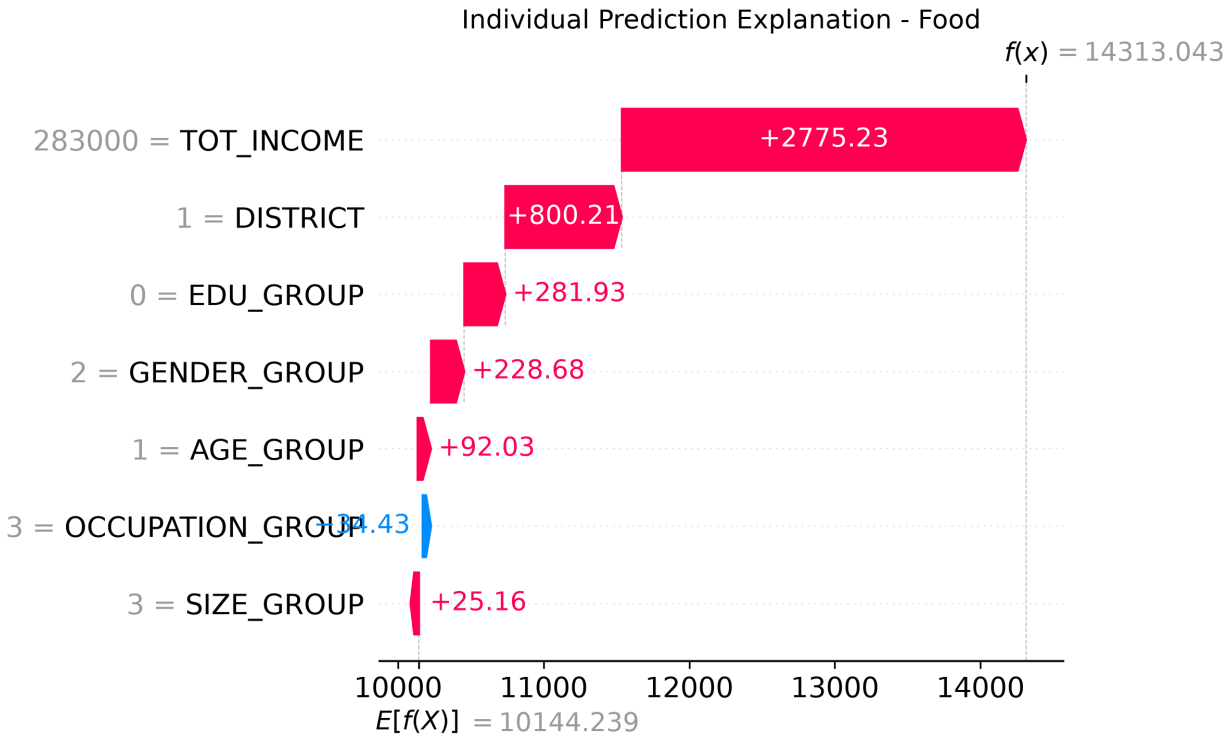
Rmse Comparison



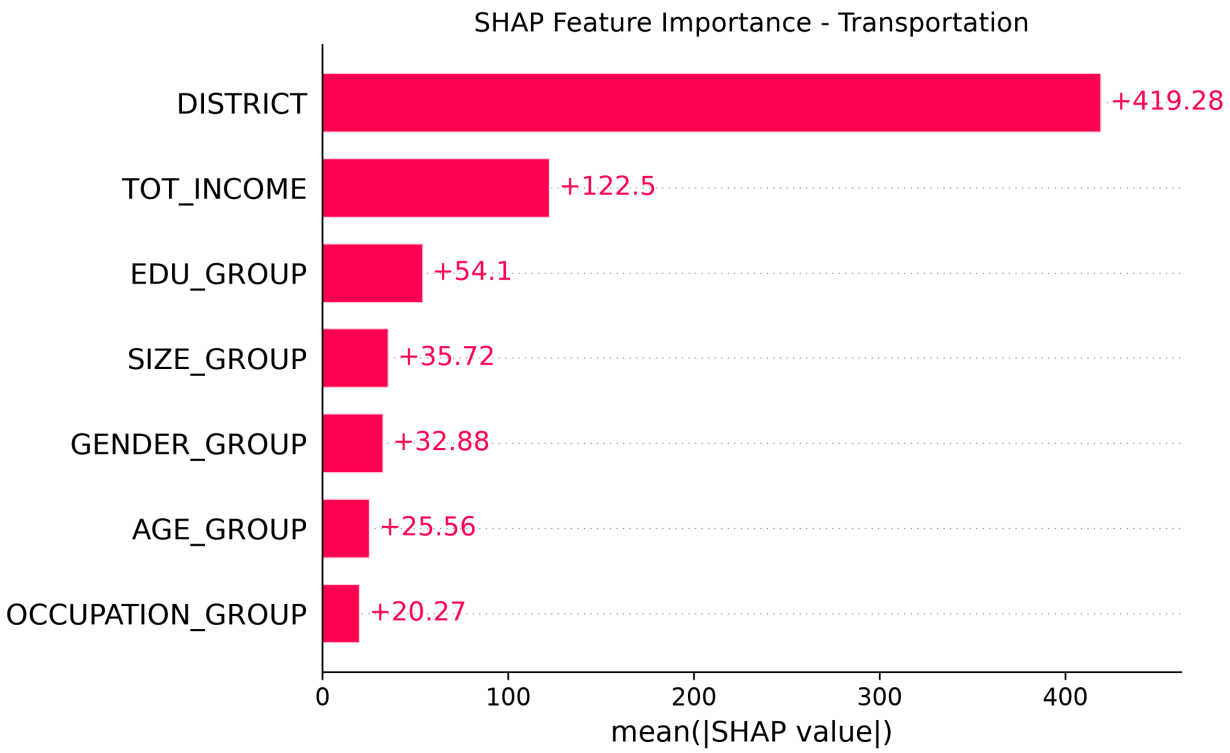
M Exp Food Shap Bar



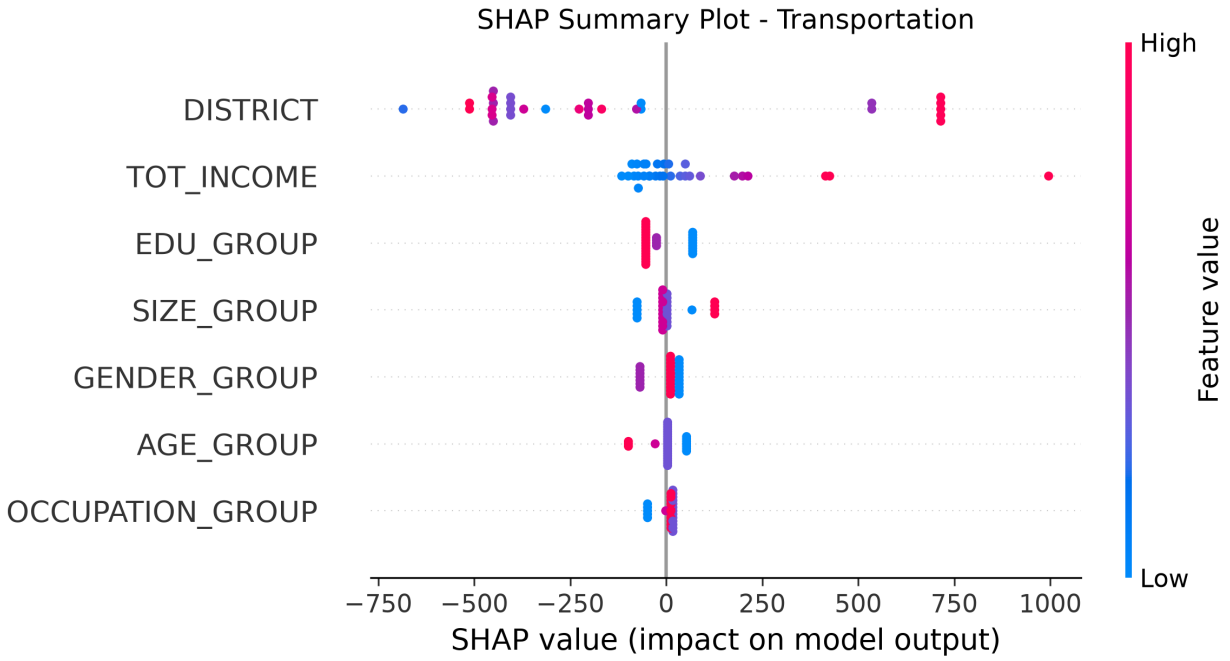
M Exp Food Shap Summary



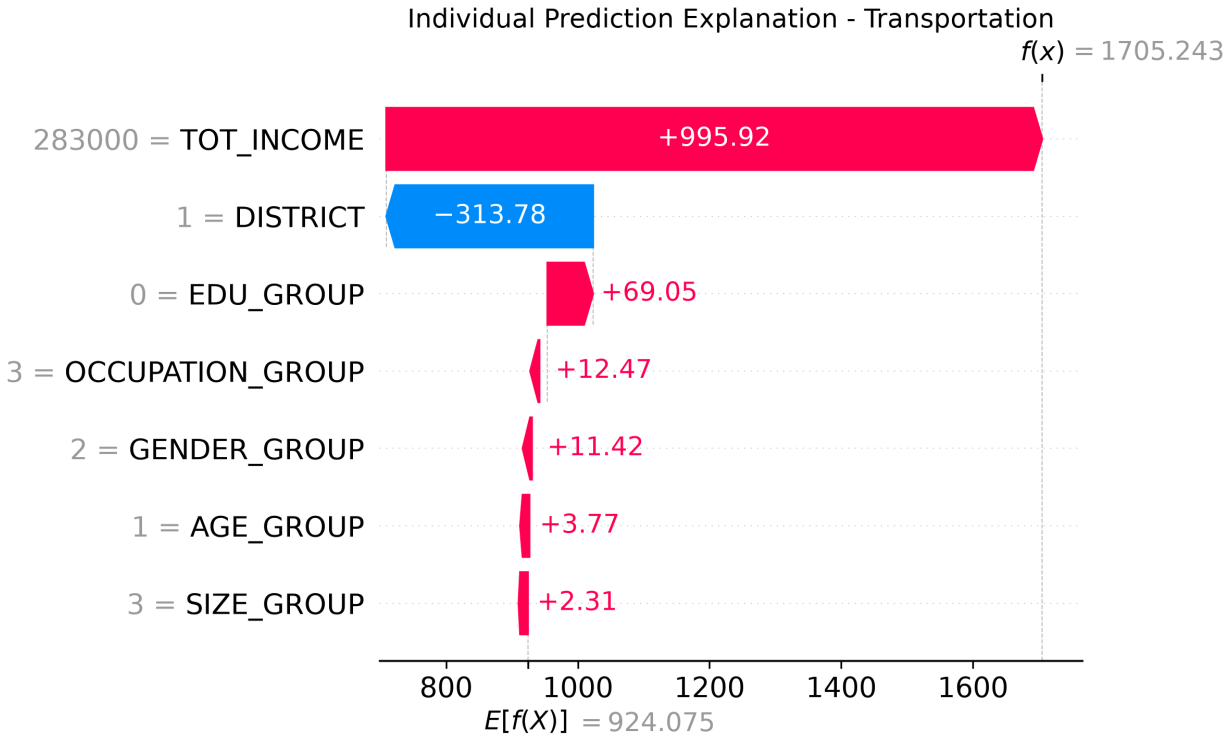
M Exp Food Shap Waterfall



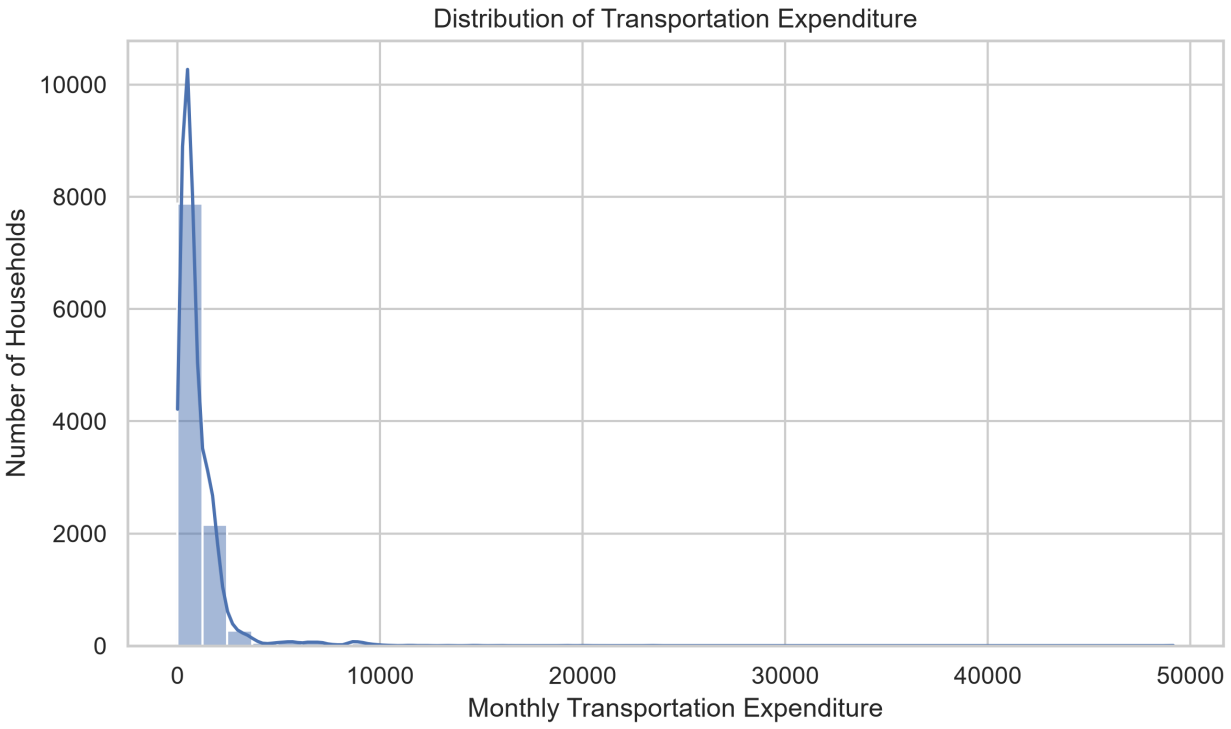
M Exp Transport Shap Bar



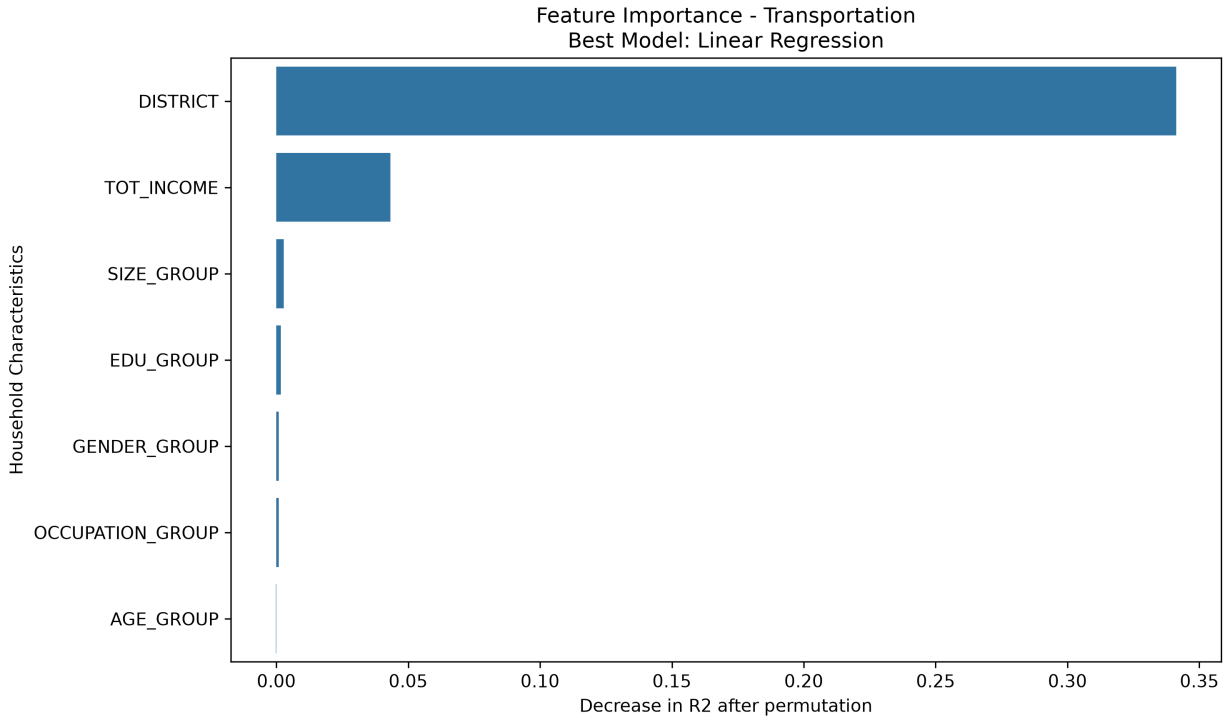
M Exp Transport Shap Summary



M Exp Transport Shap Waterfall



Transport Distribution



Transportation Feature Importance

14. RESULTS AND DISCUSSION

Food Expenditure

The food expenditure model achieved an R^2 score of 0.5816, with an MAE of approximately ■1,800 and RMSE of approximately ■2,650. This indicates that the model explains around 58.16% of the variation in food expenditure on the test dataset.

Transportation Expenditure

The transportation expenditure model achieved an R^2 score of 0.2542, with an MAE of approximately ■526 and RMSE of approximately ■1,098. The relatively lower R^2 indicates that transportation expenditure is more difficult to predict using the selected features.

Interpretation

The difference in predictive performance suggests that food and transportation expenditure may be influenced by different patterns of household behaviour. Additional variables and more advanced modelling approaches may improve predictive performance.

15. LIMITATIONS OF THE STUDY

1. The analysis is based on the variables available in the dataset.

2. The predictive performance depends on data quality and feature selection.

3. Transportation expenditure shows relatively limited predictive accuracy.

4. District-level patterns may be influenced by unequal sample sizes.

5. Machine learning associations should not be interpreted as causal relationships.

6. Further validation on independent datasets is required.

16. FUTURE SCOPE

1. Hyperparameter tuning using cross-validation.

2. Testing advanced models such as XGBoost and CatBoost.

3. Incorporating additional socioeconomic variables.

4. Using larger and more geographically representative datasets.

5. Applying explainable AI techniques for improved interpretation.

6. Developing an interactive prediction dashboard.

7. Testing model generalisation on independent household datasets.

17. CONCLUSION

This project demonstrates the application of machine learning to household food and transportation expenditure analysis in Maharashtra. The workflow combines data preprocessing, exploratory data analysis, statistical analysis, predictive modelling, feature importance and model error evaluation.

The food expenditure model achieved better predictive performance than the transportation expenditure model. The results demonstrate the usefulness of socioeconomic and demographic information for understanding household consumption patterns, while also highlighting the limitations of the current predictive models.

Further improvements through advanced model optimisation, additional features and independent validation may strengthen the reliability and generalisability of the analysis.

18. REFERENCES

1. Scikit-learn Documentation: https://scikit-learn.org/

2. Pandas Documentation: https://pandas.pydata.org/

3. NumPy Documentation: https://numpy.org/

4. SHAP Documentation: https://shap.readthedocs.io/

5. ReportLab Documentation: https://docs.reportlab.com/